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Next item →

1. In a target-tracking application, a Kalman filter can be one component of the solution. According to this lesson, what are some other possible components? Select all that apply.

1 / 1 point

☒ A method to associate targets with the correct Kalman filters tracking those targets.

☒ Correct
Yes. This is correct.

☒ A method to detect the presence of one or more targets in the sensor field.

☒ Correct
Yes. This is correct.

☒ A method to "coast" a target track when the target disappears from the sensor field.

☒ Correct
Yes. This is correct.

☐ A method to adjust the Ω parameter of the CT model to prepare for a possible future target moving in a nearly circular path.

2. Suppose that we choose to track targets using NCV models. A candidate track has passed the persistence test and we desire to initiate the execution of a Kalman filter to track the target.

1 / 1 point

The most recent position measurements have been: $z_0 = 4$, $z_{-1} = 1$, $z_{-2} = -2$, and $z_{-3} = -6$. If $\Delta t = 0.1$ seconds, what is the velocity estimate to use for $\hat{x}_0$ when initializing the filter?

30

☒ Correct
Yes. This is correct.

3. Suppose that we have detected exceedances in a 3×3 frame of data (the "track gate" comprises this entire frame):

1 / 1 point

$E_{m,n}$	$m = 1$	$m = 2$	$m = 3$
$n = 1$	10	3	0
$n = 2$	2	1	0
$n = 3$	0	0	0

What is the centroid of these exceedances in the horizontal direction?

1.25

☒ Correct
Yes. That is correct.